

Research on Building Urban Safety Emergency Response System by Computer Artificial Intelligence Automatic Control Technology

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Abstract—With the acceleration of urbanization and the increasing complexity of social risks, the construction of efficient and intelligent urban safety emergency response system has become a hotspot of current research. This paper discusses how to use computer technology, artificial intelligence and automatic control technology to construct this system. This paper reviews the relevant research background, points out the shortcomings of traditional emergency response systems, and emphasizes the importance of using advanced technologies to improve emergency response capabilities. This paper introduces a UAV-based urban emergency rescue decision support system model in detail. The model uses UAV as hardware carrier, combined with Microsoft Visual Studio 2010 and Microsoft SQL Server 2005 software platform to build a complete system architecture. Through the integration of geographic information system (GIS), UAV remote sensing technology, image recognition and other innovative technologies, the rapid acquisition and processing of accident site information, the formulation and evaluation of rescue plans, and the auxiliary management of rescue decisions are realized. The UAV-based urban emergency rescue decision support system model proposed in this paper has significant practical value and broad application prospects. Through research and practice, this paper successfully constructs a set of urban safety emergency response system model based on computer technology, artificial intelligence and automatic control technology. The computer model system not only improves the efficiency and accuracy of emergency response, but also provides scientific decision support for decision makers. The research results of this paper are of great significance for improving the level of urban safety emergency management and ensuring public safety.

Keywords—Automatic control technology, urban safety emergency response system, artificial intelligence, driverless, public security incidents, emergency rescue

I. INTRODUCTION

China is facing the serious challenge of public emergency. How to quickly detect the problems in the

emergency, so as to effectively improve the level of emergency rescue, is the urgent problem to be solved by the emergency rescue agencies and management departments. At present, according to the actual needs, a lot of research has been done on the emergency rescue system, training system and simulation and exercise system. Some researchers reproduce the actual earthquake scene through actual earthquake simulation, rapid rendering and other methods, and apply it to the actual earthquake disaster emergency drill. Some scholars built a GIS based airport emergency relief spatial database, realized the digital management of airport emergency materials, and constructed the corresponding emergency materials allocation mode to seek the optimal solution [1]. According to the characteristics and needs of enterprise emergency relief work, some scholars discussed and analyzed the architecture, design principles and main functional modules of the emergency relief command system, so as to achieve automatic monitoring and management of emergency materials, emergency basic information, emergency communication and emergency assistance decision-making. However, these systems have some shortcomings, such as relatively simple function, lack of three-dimensional cooperation and combat effectiveness [2]. Therefore, this project intends to establish a set of urban unattended emergency rescue plan based on artificial intelligence to realize real-time visual information of the accident scene through visualization, assist the headquarters to carry out synchronous analysis of the scene situation, and thus realize the optimal choice of emergency rescue path and medical evacuation path. In this way, the science and efficiency of emergency rescue are greatly improved, and the harm of accidents is greatly reduced [3]. The research results of this project have important practical significance for improving the science and efficiency of urban emergency relief work in China.

II. SYSTEM REQUIREMENTS AND FUNCTION ANALYSIS

A. System requirement analysis

In order to improve the emergency response ability, emergency rescue efficiency and emergency medical rescue ability, accurately understand and properly deal with all kinds of emergencies, strengthen the monitoring of important danger sources in public areas, and realize the "seamless" connection between the accident site, rescue command center and emergency resource scheduling [4]. Emergency rescue decision support system should have the functions of accident site information monitoring, emergency decision support, rescue information summary management, etc., so that the emergency rescue system of emergency public safety accidents can be improved, so as to realize that in the event of major emergency public safety accidents, various rescue forces can quickly, accurately and

high-quality emergency rescue and minimize casualties. Reduce the economic losses caused by accidents and ensure the safety of the people and property concerned.

B. System function analysis

A city emergency decision support system with artificial intelligence as the core is established based on the existing facilities and data [5]. The system includes four parts: public accident site information management, emergency plan design and evaluation management, emergency decision support management, data management and background management. Its functional structure is shown in Figure 1 (The image is quoted in Aiding First Incident Responders Using a Decision Support System Based on Live Drone Feeds: 19th International Symposium, KSS 2018, Tokyo, Japan, November 25-27, 2018, Proceedings).

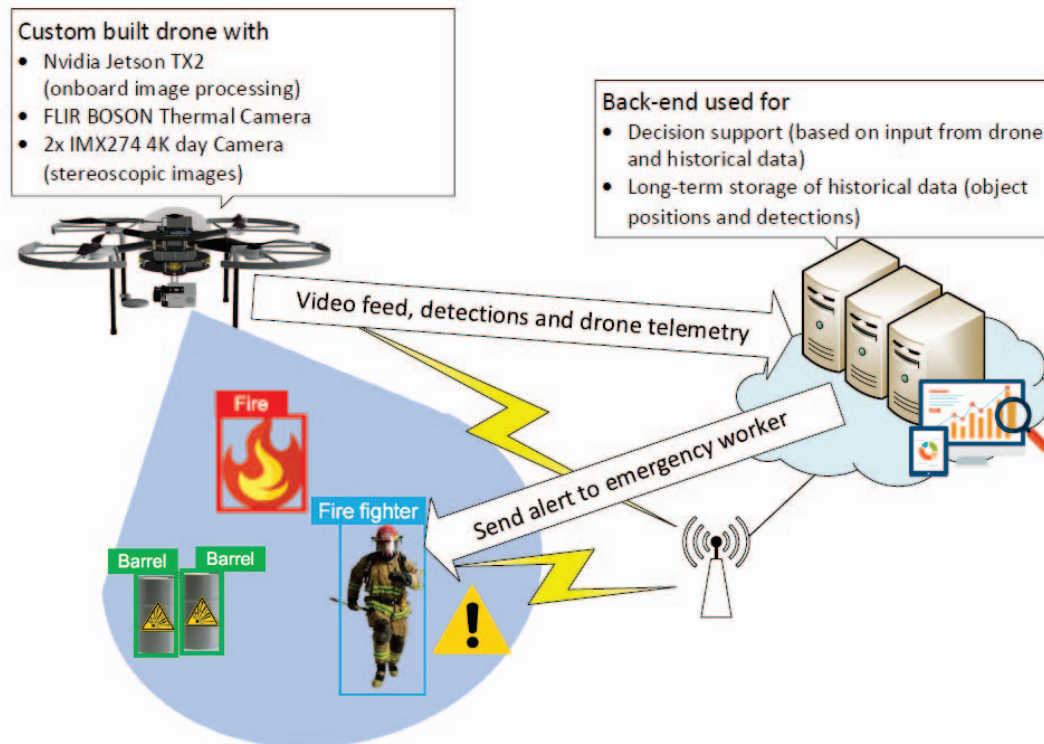


Fig. 1. Functional block diagram of decision support system with AI technology applied in UAV urban emergency disaster relief

III. URBAN SAFETY EMERGENCY RESPONSE SYSTEM DESIGN

A. System technical architecture

This topic takes "unmanned aerial vehicle" as the research object. Take "System function analysis" as the main line. The system realizes the function analysis, the design of the technical structure, the design of the database, the writing of the code and the completion of the function. The system is based on Microsoft VideStudio2010 and uses Microsoft SQLServer2005 to construct the information base of the system. At the same time, the assembled UAV hardware is integrated with the self-developed software to achieve its functional completeness, and the effectiveness of the scheme is verified through multiple tests and amendments [6]. According to the requirements of the system and the overall functional architecture, it is divided

into application presentation layer, core function layer and data service layer.

1) Use the presentation layer

Its responsibility is to provide services for various business application operations of users, and it can make modular calls to various application operations needed in the disaster medical rescue decision system, including visual user interface, interactive operation and the view of each function module [7]. It can carry out on-site information collection, casualty information identification, drug supply, medical evacuation and rescue decision-making, so as to realize the management of the whole process of public safety emergencies, such as monitoring, monitoring, early warning, decision-making, rescue and evaluation.

2) Core layer

It is an artificial intelligence-based, unattended urban emergency command and support system. Its work includes: fusion processing of information at the data service level, construction of complete data sets, and completion of various functional tasks [8]. The main contents include geographical location information collection, decision-making consultation, injury assessment of the wounded, drug support program formulation, scene shooting by UAV, individual information analysis of the wounded, and data management and maintenance.

3) Data Service layer

This paper focuses on the management of facilities and information in emergency situations. Device management

refers to the management of monitoring and communication devices, such as GPS devices, video devices, mobile terminals, and voice intercom, which are used to transmit commands and collect various information in emergency situations [9]. Data processing includes spatial data and functional data, which realize data collection, processing, analysis and storage, and provide key data support for emergency medical aid work at all levels, such as emergency medical aid and auxiliary decision-making. Figure 2 shows the technical architecture of the system (image cited in UAV Framework for Autonomous Onboard Navigation and People/Object Detection in Cluttered Indoor Environments).

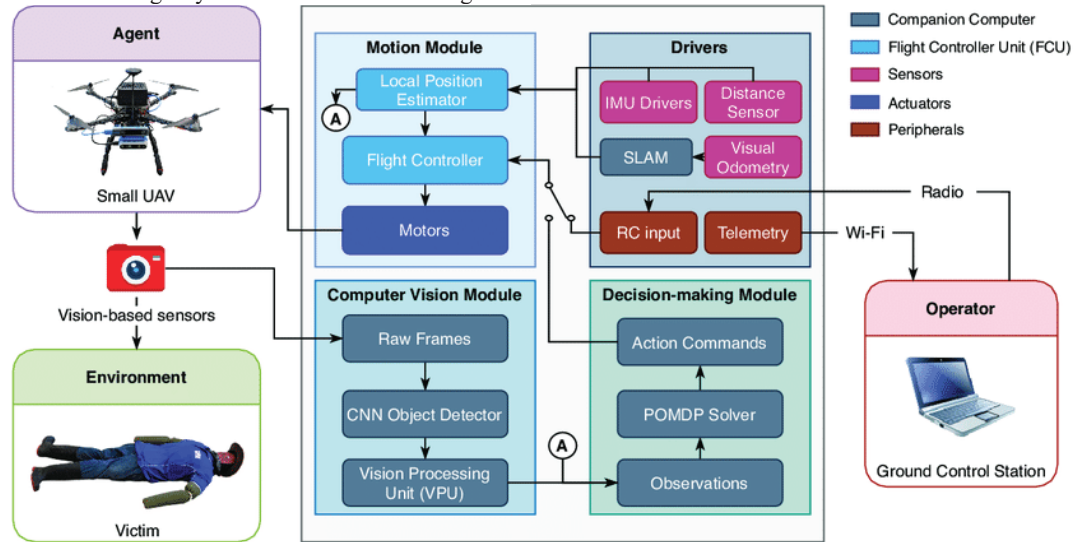


Fig. 2. Technical architecture of application of AI technology support in UAV urban emergency disaster relief

B. System function design

The public event scene information processing system is mainly responsible for collecting and recording the identity and physiological characteristics of the injured. The formulation, evaluation and management of the rescue plan, the evaluation of the vital characteristics of the injured, the classification of the injured, the decision of the treatment and evacuation order of the injured, and the integration and analysis of the collected information, and the final formation of a set of emergency rescue plan. The assistance

management model of emergency rescue formulated the plan of medicine, materials and rescue number needed for rescue, calculated the number of injured people and the number of injured people, so as to determine the type and number of rescue vehicles and equipment [10]. Data processing and back-end maintenance functions: complete all kinds of data query, modification, update, maintenance, summary and other functions of the system. Figure 3 shows the workflow of the System (image cited in Co-TS: Design and Implementation of a 2-UAV Cooperative Transportation System).

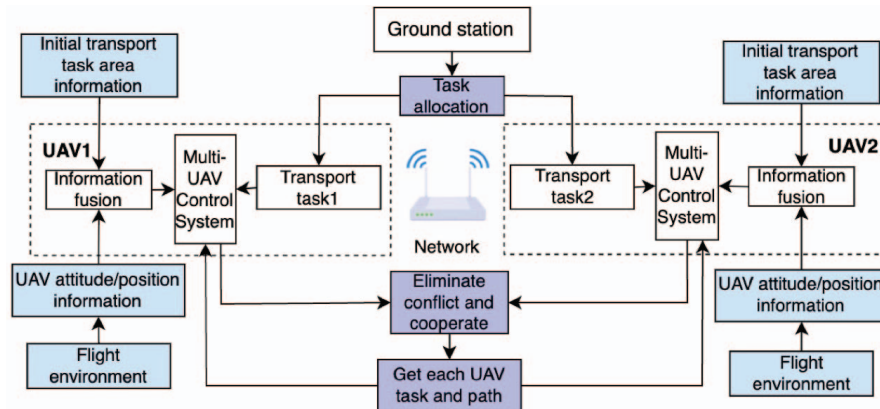


Fig. 3. Operation flow of decision support system using AI technology in UAV urban emergency disaster relief

C. Data structure design

The basic idea of the system is: reasonable organization, clear structure, less redundancy, easy maintenance, easy operation and strong expansibility. This project takes large-scale public health emergencies in China as the research object, and constructs a set of unmanned emergency rescue

plan suitable for China's national conditions [11]. Reasonable data organization and database structure design are very important to ensure the normal operation of the system. The system uses SQL Server 2005 database to abstract the function of the system. The E-R diagram of its logical data model is shown in Figure 4 (the picture is cited in Drones 2023, 7(3), 214).

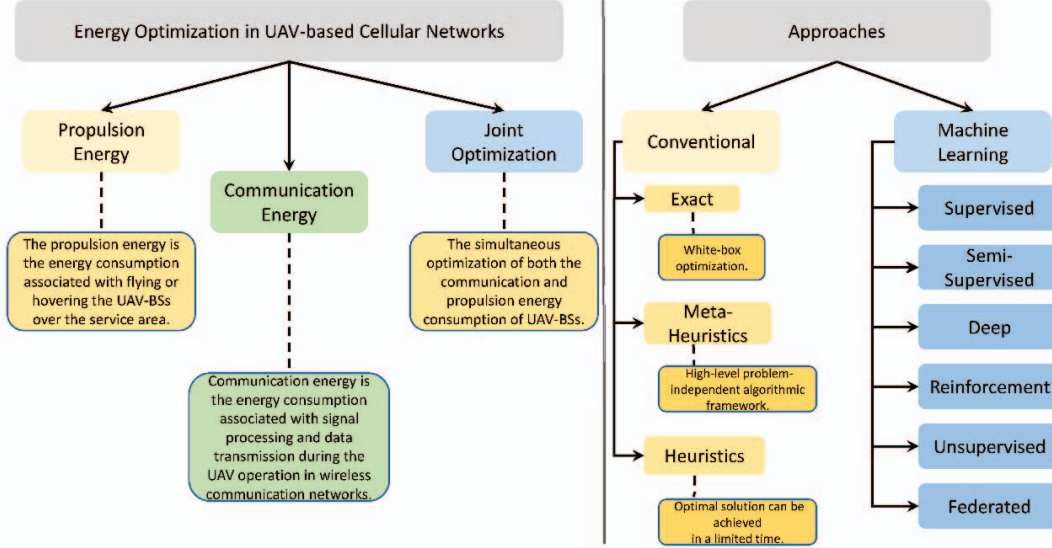


Fig. 4. E-R diagram of AI-supported decision support system in UAV urban emergency disaster relief

D. Database Table Design

In view of the geographical location, injured personnel, medicine and equipment information existing in the emergency disaster relief of UAV city, it must be digitized and the corresponding database form must be established. A new UAV-oriented urban emergency response plan support system is proposed [12]. The data field should be included in the database table, the data type, and whether null is allowed. In this system, the medical evacuation table, the hospital treatment table, the public emergency classification table, the personnel information table, the disaster relief decision-making table, and the drug supply guarantee table have been established.

IV. SYSTEM ALGORITHM DESIGN

A. Isomorphic SPN model Markov chain

The chain state is a map of each identity of the SPN. The specific transformation process is as follows:

1) The reachability map is obtained according to the SPN model, the reachability identification set is determined, and the identification state U_i of each library is obtained.

2) According to the actual measurement, the average implementation rate γ_i of transition k_i is obtained, that is, the average implementation times per unit time when the transition can be implemented. The implementation transition k_i marked on each arc is replaced by γ_i , and the isomorphic Markov chain of the SPN model can be obtained.

B. Index selection and calculation

This paper mainly analyzes the main executive departments, task completion and emergency response time of subway emergency response process, so the busy probability of library, transition utilization rate and average execution time of system are selected for calculation and analysis.

1) Average implementation

$$\begin{cases} \gamma_i = \frac{1}{K_i} & K_i \neq 0 \\ 0 & K_i = 0 \end{cases} \quad (1)$$

Where: γ_i is the transition k_i ; K_i is the execution time of the link represented by transition k_i . If $K_i = 0$, it indicates that the link is not involved in the emergency process.

2) Stability probability of reachable mark

First, all equilibrium equations of state are listed according to isomorphic Markov chains, and for any identified $U_i \in [U_0 >]$, all $U_c, U_j \in [U_0 >]$, and $U_i[k_j > U_c, U_j[k_j > U_i]$, equation (2) is given:

$$(\sum_c \gamma_c) v_i = \sum_j (\gamma_j v_j) \quad (2)$$

Where: γ_c, γ_j is the average implementation rate of transition k_c and k_j respectively; v_i, v_j is the stability probability that identifies U_i and U_j , respectively.

Take $n-1$ equilibrium equation of state and combine it with $\sum_i v_i = 1$ to calculate the stability probability v_i for each reachable identifier.

3) Library busy probability

Let $Q[U(q_i) = c]$ be the probability that library q_i contains c tags, then the busy probability of the library is shown in formula (3):

$$Q[U(q_i) = c] = \sum Q[U_d], c \geq 1 \quad (3)$$

Where: $Q[U_d]$ is the stability probability of U_d ; U_d indicates the reachable identifier.

4) Transition utilization

Define $Y(k_i)$ as the transition utilization rate, then $Y(k_i)$ is equal to the sum of the stable probabilities of transforming k_i into the enable transition mark, as shown in equation (4):

$$Y(k_i) = \sum_{U \in W} Q(U) \quad (4)$$

Where: W is all the reachable identifier sets enabled by k_i .

5) Transition marked flow rate

Marked flow velocity of transition k_i refers to the average number of marks $S(k, d)$ that flow into the post position q of k_i in unit time, as shown in equation (5):

$$S(k, d) = F(k, q) \times Y(k) \times \gamma \quad (5)$$

Where: k refers to change; d refers to the warehouse; $F(k, q)$ is the marker value that flows into the post position q through transition k_i ; γ is the transition.

6) Average System Execution Time

According to the Little rule and the principle of balance, the average execution time of the system is shown in equation (6):

$$T = O / S(k, d) \quad (6)$$

Where: O represents the average number of marks contained in the location set at steady state.

V. CONCLUSION

This topic uses the UAV as the carrier, equipped with vital signs detection device, to realize the information collection, function realization, data processing and rescue plan formulation for the injured. By drawing lessons from foreign public health emergency rescue and service mechanisms, the equipment and rescue efficiency of public security organs and emergency rescue agencies are improved, and the cooperation and cooperation between various departments are strengthened. And in the actual work constantly adjust and improve, in order to effectively provide effective auxiliary decision support for emergency rescue, improve emergency response speed, rescue ability and decision-making ability.

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